

Claude Code

A field guide for data scientists

What We Will Cover

- 01 What Claude Code is
- 02 How it differs from chat
- 03 Getting set up
- 04 Everyday data science workflows
- 05 Habits that make it shine



Claude Code

A terminal-based AI agent that reads your project, writes and runs code, inspects the results, and iterates until the job is done.

Chat vs Agent

Chat

You ferry code around

vs

Agent

It works in your repo

What It Can Do in Your Project

Explore

Read files, schemas, and notebooks.

Execute

Run scripts, queries, and tests.

Iterate

Check outputs and fix its own errors.

Set Up in Three Steps

01 Install

`npm install -g @anthropic-ai/claude-code`

02 Open your project

cd into the repo and run claude.

03 Ask for an analysis

Describe the outcome; it plans the code.



a first session

```
$ cd sales-analysis && claude
```

```
> Profile sales.csv: nulls, dtypes,  
  outliers. Save it to profile.md.
```

```
Reading sales.csv (1.2M rows)...
```

```
Running profiling script...
```

```
Wrote profile.md: 3 findings.
```

Everyday Workflows

EDA

Profile new data in minutes.

Cleaning

Fix types, nulls, duplicates.

Modeling

Fast first baselines.

Reporting

Plots and write-ups.

How It Works on a Task

01

Explore

Scans the data and code.

02

Plan

Proposes an approach.

03

Execute

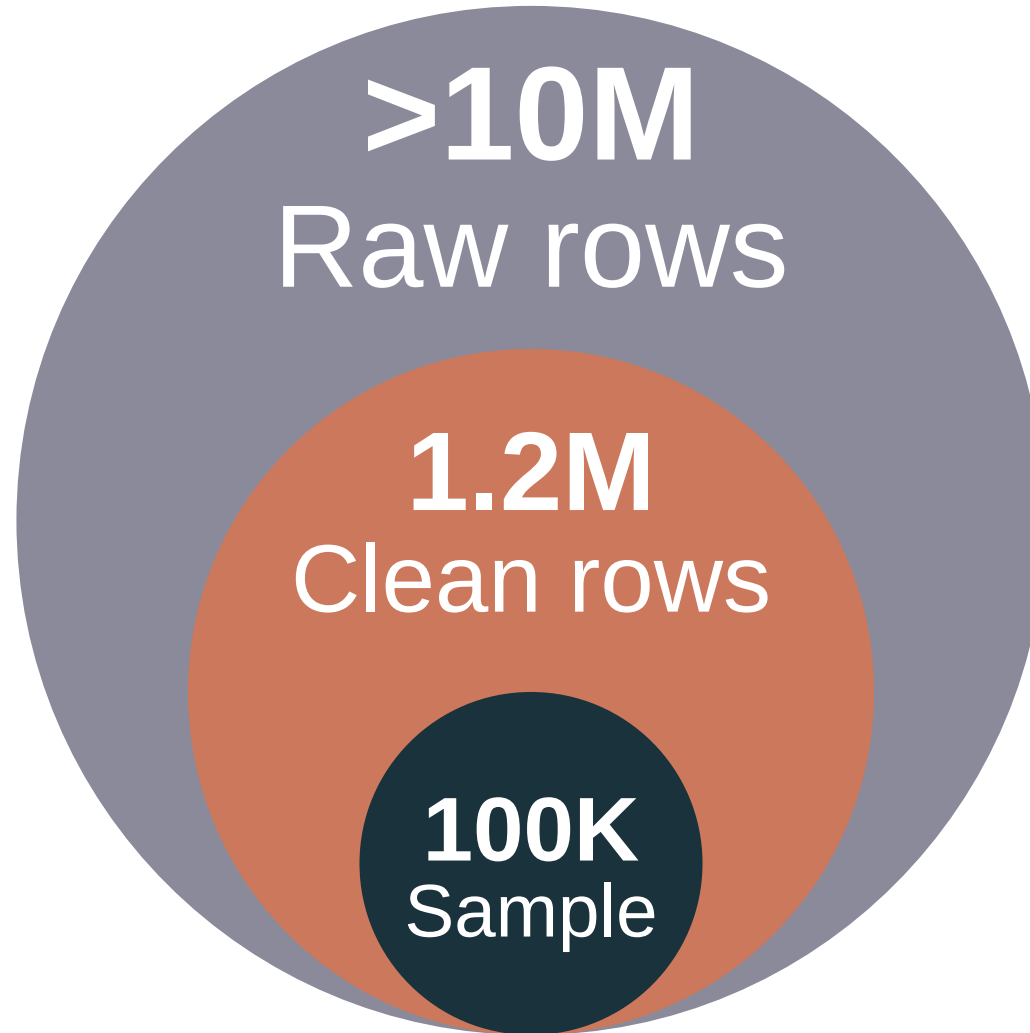
Writes and runs the code.

04

Verify

Checks results, then iterates.

One Prompt, a Whole Pipeline



A Typical Day, Reworked

Before

Boilerplate pandas by hand

Stack Overflow detours

Stale, undocumented
notebooks

With Claude Code

Describe the outcome
instead

Errors debugged in place

Analysis committed with
notes

Commands Worth Knowing

Command	What it does
<code>claude</code>	Start a session
<code>/init</code>	Write CLAUDE.md
<code>/compact</code>	Trim the context
<code>claude -p</code>	One-shot mode

Habits That Make It Shine

Keep a CLAUDE.md with data caveats.

Ask for a plan before a big analysis.

Have it verify outputs, not just run.

Commit small, reviewable steps.



01

One Claude, Many Surfaces

Same Engine, Everywhere

One engine drives the terminal, IDE, desktop, and web.

Your CLAUDE.md, settings, and MCP servers come along.

Start on your laptop, pick it up on your phone.

Three Ways to Run It

Terminal

Web

Desktop

Runs on

Your
machine

The cloud

Your
machine

Setup

Install CLI

None

Install app

Best for

Full control

On the go

Visual diffs

In the Terminal

The full CLI — edit files and run commands.

Runs locally with direct access to your tools.

Install with one script, then just run claude.

On the Internet

Cloud-run

Runs at `claude.ai/code` on Anthropic's cloud.

Isolated

Each task gets its own sandbox VM.

Persistent

Keeps running after you close the tab.

Claude Desktop

A standalone app outside your IDE or terminal.

Review diffs and run sessions side by side.

Schedule repeat tasks and start cloud runs.



02

Drive It From Your Phone

Connect GitHub

01 Sign in

Open `claude.ai/code` and log in.

02 Link GitHub

Grant access to the repos you choose.

03 Pick a repo

Claude clones it into a cloud sandbox.

Start a Task Anywhere

01 Describe it

Type the task in the Claude mobile app.

02 Let it run

Work continues in the background.

03 Review

Approve the branch Claude pushes.

Remote Control

Step away and keep working from your phone.

Continue a live session from any browser.

Pull a cloud task back with `claude --teleport`.



move between surfaces

```
$ claude --cloud
```

Runs this task **in** the cloud.

> Pick it up later, anywhere:

```
$ claude --teleport
```

One session follows you across
laptop, phone, **and** browser.



03

MCP: Plug In Your Tools



MCP

An open standard that connects AI tools to your data sources and other systems.

What MCP Unlocks

Read design docs straight from Google Drive.

Update tickets in Jira, pull data from Slack.

Query your own database and internal APIs.

Servers You Can Connect

Server	What Claude can do
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Drive	Read design docs
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Jira	Update tickets
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Slack	Pull conversations
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Postgres	Query your data
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04

Make Claude Yours



CLAUDE.md

A file in your project root that Claude reads at the start of every session.

What Goes in CLAUDE.md

Coding standards and preferred libraries.

Architecture decisions and key commands.

Review checklists and data caveats.



Skills

Reusable folders of instructions, each a SKILL.md, that Claude loads on demand.

CLAUDE.md vs Skills

CLAUDE.md

Always in context

Project-wide rules

One file at the root

Skills

Loaded on demand

A packaged workflow

Shared across projects



**What would you analyze if
the boilerplate wrote itself?**

Start Today

Open a terminal and ask